

DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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Abstract—Physics-based humanoid motion imitation must simultaneously satisfy several heterogeneous objectives, including global root motion, articulated pose, and velocities, keeping them consistent with a synchronized reference trajectory. Handcrafted rewards combine these objectives with manually designed weights and scales, whereas adversarial imitation can learn a unified imitation signal from data through a discriminator, reducing the reliance on explicit reward weights. For differential discriminators that take the reference–policy differential as input, the sensitivity of the score function and its numerical range still have to be controlled; otherwise, changes in the residual distribution across motions can make the same regularization setting produce different effective learning signals. We propose DARE, a differential adversarial reward method that requires no motion-specific coefficient and controls the geometry of the score function and the score scale separately. DARE maps each semantic residual block through its own map into a representation of a common latent width, with an independent spectrally normalized gain bound imposed on each map, thereby bounding the score change relative to the unique zero-error anchor. DARE then performs a one-shot closed-form score calibration according to the response window of the reward transform, with the global scaling factor determined automatically from the residual distribution observed during training. We evaluate DARE with a single configuration on six humanoid skills with diverse dynamical characteristics and contact patterns. DARE attains the lowest cross-motion macro-average error on all four geometric tracking measures, the foot progress closest to the reference, and the lowest lateral drift; the resulting controllers are further deployed on **[TODO: robot platform]**.

I. INTRODUCTION

Physics-based humanoid motion imitation requires the controller to reproduce a prescribed reference motion under dynamic constraints, actuator limits, and complex contact conditions. Whether an imitation succeeds is not a single geometric question. The root must translate and turn as the reference does, the articulated body must maintain consistent poses, joint and root velocities must stay synchronized with the reference motion, and contacts must occur at the right moments. Because these objectives have different physical units, numerical ranges, and dynamical characteristics, and are coupled through the robot dynamics, motion imitation is inherently a multi-objective control problem.

Traditional methods design a separate reward term for each type of tracking error and combine them into a scalar reward through manually chosen weights and scales. A representative form is

$$r_t^{\text{hand}} = \sum_{g=1}^G w_g \exp\left[-\alpha_g \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2^2\right], \quad (1)$$

where ϕ_g extracts the g -th motion feature—joint rotations, root position, velocities, or end-effector positions—and s_t and $\hat{s}_t$ denote the current simulated state and the synchronized reference state. The weights w_g decide the relative importance among tracking objectives, and the scales α_g fix the numerical range of each error in the reward. The difficulty is that typical error magnitudes, dynamical difficulty, and the relative importance of the objectives change with the motion, so a coefficient set that serves one skill need not serve another; in practice, reward design is re-tuned whenever the motion changes.

Adversarial imitation offers a way to reduce manual reward design. Instead of declaring in advance how many meters of root-position error are worth how many radians of rotation error, a discriminator learns a unified imitation signal directly from data. AMP and related methods learn rewards that describe motion style and plausibility by discriminating whether short state transitions come from the reference distribution [5]. For motion tracking that must strictly follow a synchronized reference trajectory, ADD further feeds the reference–policy state differential to the discriminator and asks whether it is close to the ideal zero differential [6]. The aggregation among tracking objectives can then be learned by the discriminator, instead of specifying the per-term reward weights and scales of (1) by hand.

Differential adversarial rewards, however, still face an important issue. Their ideal sample is always a single point, the zero differential. Without a suitable constraint, the discriminator easily becomes too sharp: it assigns a high score to the origin and compresses the many nonzero residuals produced by the policy into a nearby low-score region. Even if classification succeeds, the discriminator may not provide an effective continuous learning signal for the policy. Existing methods constrain the discriminator through regularization, but one regularizer simultaneously affects two distinct properties. On the one hand, it controls the discriminator’s local sensitivity, that is, how fast the score changes with the input residual. On the other hand, it also affects the numerical range in which the final scores live, and this range in turn determines whether the policy reward operates in an effective interval. As the residual distribution changes across motions, the same regularization setting can correspond to different sensitivities and different score scales, producing motion-dependent effective regularization requirements.

The core idea of DARE is to treat these two issues separately rather than control them with one motion-dependent coefficient. Consider first the critic’s sensitivity to different tracking objectives. For any motion and any phase, perfect im-

itation always corresponds to the same zero differential $\Delta = 0$, which we call the anchor. Meanwhile, the differential in motion tracking naturally decomposes into semantic blocks with clear physical meaning, such as root motion, body pose, and velocities. Different tracking objectives occupy residual blocks of different dimensions, which can couple representation size to critic sensitivity. DARE first maps each semantic block through its own spectrally normalized map to a common-width latent representation before shared aggregation; the shared critic then learns the relations among objectives from the residuals that actually occur during training. The concrete construction and the corresponding Lipschitz properties are given in Section IV-B.

Constraining the critic’s sensitivity alone does not resolve the score scale. A Lipschitz bound controls how rapidly the score may change, but not the numerical range actually used by the trained critic. DARE therefore measures the anchor-to-typical-residual score gap once after normalization stabilizes and uses a closed-form global rescaling tied to the response range of the reward transform. All motions use exactly the same calibration rule; the only quantity that changes across motions is the residual statistic measured automatically from the training data. The critic’s geometry and score scale thus no longer depend on the same hand-tuned parameter.

The main contributions are threefold. First, we identify that the regularizer in existing differential adversarial rewards simultaneously affects the critic’s local geometry and its score scale, and that this coupling can make different motions require different effective regularization settings. Second, we propose DARE, which handles the semantic residual blocks separately through a factorized spectrally normalized critic and sets the score scale automatically by a one-shot anchor calibration, removing the need to re-specify reward or regularization coefficients for each motion. Third, we evaluate one DARE configuration on six skills with different motion durations, contact patterns, and dynamical characteristics, and deploy the resulting controllers on **[TODO: robot platform]**.

II. RELATED WORK

Physics-based motion tracking and learned imitation objectives. DeepMimic combines pose, velocity, root, and end-effector tracking terms for physics-based skills [1]; subsequent work improves scalability and motion coverage through architectures, representations, and data [2], [3]. Adversarial imitation provides a learned alternative to a manually written reward: GAIL matches expert and policy occupancies [4], while AMP learns a motion prior by discriminating short state transitions [5]. ADD addresses reference-conditioned tracking by presenting the reference–policy differential to an adversarial discriminator and learning a nonlinear aggregation of tracking objectives without component-wise reward weights [6].

Multi-objective learning and objective aggregation. Competing objectives are commonly combined by weighted scalarization, Pareto optimization, or adaptive loss weighting [10], [11]. These approaches differ in whether preferences

are specified explicitly or adapted from objective statistics. Motion tracking adds a representation issue: one physical objective can occupy a feature block with many more coordinates than another, even when both are conceptually single objectives.

Regularization of adversarial critics. Adversarial critics are regularized with gradient penalties or spectral normalization to control sensitivity and improve optimization [7], [8]. Gradient penalties constrain sampled input derivatives, whereas spectral normalization bounds the gain of individual network maps by construction. For non-Wasserstein adversarial losses, the numerical score range also determines the operating regime of the loss and its reward transform [9]. Geometry and score calibration are therefore related but distinct considerations when a critic supplies the learning signal.

III. BACKGROUND

A. Reference-Conditioned Motion Tracking

In reference-conditioned humanoid motion imitation, the controller must keep the executed motion continuously aligned with a time-synchronized reference trajectory. Let s_t and $\hat{s}_t$ denote the current state and the reference state at time t . Tracking quality is jointly measured by several physical quantities, including root position and orientation, articulated body pose, and motion velocities. We organize these quantities into G semantic feature groups: $\phi_g(s_t)$ extracts the feature vector of group g from the current state, and $\phi_g(\hat{s}_t)$ its counterpart from the reference state. Different groups may have different dimensions and physical units—positions are measured in meters, while rotations and velocities have their own representations.

B. Handcrafted Objective Scalarization

Traditional methods first compress each tracking error into a scalar. For group g , the scalar tracking error at time t is the Euclidean distance $e_{g,t} = \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2$. The errors $e_{g,t}$ are then transformed and combined into a single policy reward. DeepMimic is a representative instance of this class: its tracking terms use weights and error scales specified before training and held fixed throughout [1]. This is a fixed scalarization of the tracking objectives: the trade-off among errors is fixed through constant coefficients chosen before training starts. Because typical error magnitudes and dynamical characteristics differ across motions, one coefficient set need not serve all motions. Learned scalarization instead leaves the trade-off unspecified in advance and learns the aggregation among objectives from the error distribution that actually occurs during training.

C. Differential Adversarial Objectives

Differential adversarial learning provides such a learned aggregation [6]. Consider n non-negative objective losses $l^1, \dots, l^n$ collected into a differential vector $\Delta = (l^1, \dots, l^n)$, each entry recording the gap between the current solution and the ideal on one objective; when all objectives are simultaneously ideal, $\Delta = 0$. A discriminator $D(\Delta)$ is trained to

separate the ideal zero differential from the nonzero differentials produced by the current policy. Since the discriminator receives the differentials of all objectives at once, it can learn a nonlinear objective aggregation function without fixed per-objective weights. Its local sensitivity to objective i is described by $\partial D / \partial l^i$: the extent to which the discriminator output responds when objective i changes slightly near the current differential. Unlike fixed scalarization weights, this local sensitivity can vary with the position in the residual space and during training. In synchronized motion tracking, the differential is no longer restricted to one scalar loss per objective but can be formed directly from the feature residual between the current and reference states. Perfect tracking again corresponds to the unique zero differential, while nonzero differentials retain the error direction and the relations among features. The policy reward is generated from the discriminator's score on the current differential, for example $r_t = -\log(1 - D(\Delta_t))$: the closer the current residual is judged to the ideal state, the higher the reward.

A peculiarity of this form is that the ideal class always concentrates on the single point $\Delta = 0$. If the discriminator is overly sensitive to input changes, it can easily separate the origin from nonzero residuals, yet fail to form a smooth and useful score among policy residuals of different quality. Differential adversarial methods therefore constrain the local sensitivity of the score function, for example with a sampled gradient penalty $\lambda \mathbb{E}_{\Delta \sim q_\pi} \|\nabla_\Delta D(\Delta)\|_2^2$, where q_π denotes the residual distribution produced by the current policy [6]. Constraining the local sensitivity, however, does not determine the score range the trained discriminator uses: since the policy reward further passes through a nonlinear transform, both properties affect the final learning signal, but they are not the same question.

IV. DARE: DIFFERENTIAL ADVERSARIAL REWARD EQUALIZATION

A. Learned Differential Reward

Motion tracking involves multiple objectives with different physical meanings. For each objective, the scalar distance $e_{g,t}$ summarizes the error magnitude and is well suited to a traditional weighted reward, but it retains only that magnitude. The raw feature differential also contains the error direction and the joint variation across coordinates. DARE therefore takes the feature differentials directly as the input of the learned reward. For semantic group g , the raw residual is $\phi_g(\hat{s}_t) - \phi_g(s_t)$. Since positions, rotations, and velocities have different units and typical ranges, we normalize every coordinate first. Let the normalizer of group g be $\mathcal{N}_g = \text{diag}(\nu_g)^{-1}$, where ν_g records the mean absolute magnitude of each coordinate's residual, estimated from residual samples in early training and fixed after the first 10^8 samples, as is standard for adversarial motion rewards. The normalization only rescales coordinates and introduces no offset, so the zero differential of perfect tracking always stays at the origin; rotations use a tangent-normal representation so that their residuals can be handled uniformly in Euclidean coordinates.

After normalization, the group residual, the full differential, and the ideal point are

$$\Delta_{g,t} = \mathcal{N}_g(\phi_g(\hat{s}_t) - \phi_g(s_t)), \quad \Delta_t = \bigoplus_{g=1}^G \Delta_{g,t}, \quad \Delta^+ = 0. \quad (2)$$

The vector $\Delta_t \in \mathbb{R}^d$, $d = \sum_g d_g$, retains the direction of each coordinate error and the structural relations formed when several coordinates vary together. Whatever the reference motion and whatever the current phase, perfectly consistent states map to the same zero differential, while the nonzero differentials produced by the policy record how the current motion deviates from the reference. During training, recently visited differentials are kept in a replay buffer and combined with the current ones into the policy residual distribution q_π , which keeps evolving with the policy.

DARE learns a scalar score function on this differential space. Let $f_\theta(\Delta)$ denote the raw scalar score of the critic, converted by the logistic sigmoid into $D_\theta(\Delta) = \sigma(f_\theta(\Delta))$, which expresses the degree to which the critic considers the differential ideal. The zero differential represents perfect tracking, and the residuals produced by the policy are drawn from q_π ; the critic is therefore trained by the following adversarial objective to distinguish the two:

$$\max_{\theta} \log D_\theta(0) + \mathbb{E}_{\Delta \sim q_\pi} [\log(1 - D_\theta(\Delta))]. \quad (3)$$

The first term raises the ideality score of the zero differential, and the second lowers the ideality score of current policy residuals. As the policy improves, the residual distribution gradually moves toward the origin, and the critic updates its score function accordingly. A function f_θ that co-evolves with training thereby aggregates multiple tracking objectives into a single scalar signal.

For optimization, we write the above binary classification objective as a balanced Bernoulli negative log-likelihood. Using $-\log \sigma(z) = \text{softplus}(-z)$ and $-\log(1 - \sigma(z)) = \text{softplus}(z)$ gives the critic loss

$$\mathcal{L}_{\text{DARE}}(\theta) = \frac{1}{2} \text{softplus}[-f_\theta(0)] + \frac{1}{2} \mathbb{E}_{\Delta \sim q_\pi} \text{softplus}[f_\theta(\Delta)]. \quad (4)$$

This objective provides the critic with only two kinds of supervision, ideal differentials and policy residuals; how root motion, body pose, and velocities jointly determine the final score is learned by f_θ from the residual distribution observed during training.

The policy then constructs its reward directly from the critic score. DARE defines

$$r_t^{\text{DARE}} = -\beta \log[1 - D_\theta(\Delta_t)] = \beta \text{softplus}[f_\theta(\Delta_t)], \quad (5)$$

where β controls the overall reward amplitude and is held fixed across all motions. The closer the critic judges the current differential to the ideal state, the higher $f_\theta(\Delta_t)$ and the larger the policy reward. The policy can thus improve root motion, body pose, and velocities simultaneously through one unified scalar reward.

This formulation recasts multi-objective tracking as learning a scalar score function in the differential space, and its

effect depends on two properties of the critic itself. First, the semantic feature groups in the differential have different dimensions, and the critic’s sensitivity to each group’s residual variation requires a clear and consistent structural constraint. Second, the scores produced by the critic are converted into the policy reward through the softplus transform, so the numerical range of the scores directly determines how strongly the reward responds to changes in tracking quality. DARE states these two properties as critic geometry and score scale, and designs for them independently.

B. Equalized Critic: Geometry by Construction

The full differential Δ consists of G semantic feature blocks, the g -th block denoted $\Delta_g \in \mathbb{R}^{d_g}$, where d_g is the feature dimension of that objective. Different objectives in motion tracking use different physical representations, so their coordinate counts usually differ. Per-coordinate normalization makes the typical numerical scales of coordinates comparable, but the overall norm of a feature block is still influenced by its dimension. When coordinates have comparable typical magnitudes, the block norm $\|\Delta_g\|_2$ generally grows with $\sqrt{d_g}$. Consequently, if all semantic blocks were concatenated and fed directly into one shared dense layer, the raw representation dimension would participate in deciding how strongly the network responds to each objective’s residual variation.

Let this shared linear map be $W = [W_1 \cdots W_G]$, where W_g acts on the g -th semantic block. The largest singular value $\sigma_{\max}(W)$ of a linear map expresses the maximum amplification it can apply to an input change, and imposing $\sigma_{\max}(W) \leq 1$ means that distances between any two points are at most preserved—a 1-Lipschitz property. For any unit output direction u , the shared map satisfies

$$\sum_{g=1}^G \|W_g^T u\|_2^2 = \|W^T u\|_2^2 \leq 1, \quad (6)$$

which shows that the semantic blocks are jointly subject to one overall gain constraint in each output direction. A shared layer can bound the maximum amplification of the whole map, but the gains of the blocks are allocated by the same matrix, so its response is coupled to the input representation itself. For semantic blocks of different dimensions, this coupling lets the representation dimension influence how sensitive the critic is to each objective’s residual variation.

DARE instead builds an independent constrained representation for each semantic block before the blocks interact. The g -th block is mapped into the latent space by

$$\psi_g(\Delta_g) = \rho(\bar{W}_g \Delta_g + b_g), \quad \bar{W}_g = \frac{W_g}{\sigma_{\max}(W_g)}, \quad (7)$$

where ρ is a 1-Lipschitz activation. Spectral normalization rescales the largest singular value of each linear map to one, so each ψ_g carries its own independent 1-Lipschitz gain bound: $\|\psi_g(\Delta_g) - \psi_g(\Delta'_g)\|_2 \leq \|\Delta_g - \Delta'_g\|_2$. All semantic blocks are mapped to the same latent width m ($W_g \in \mathbb{R}^{m \times d_g}$, $\psi_g(\Delta_g) \in \mathbb{R}^m$), giving each objective the same number of representation channels before shared aggregation.

The latent representations of the blocks are then combined by a direct sum, $\Phi_{\oplus}(\Delta) = \bigoplus_{g=1}^G \psi_g(\Delta_g)$, where $\bigoplus$ places each m -dimensional latent vector in mutually disjoint coordinates; with G semantic blocks the total dimension of $\Phi_{\oplus}(\Delta)$ is Gm . Within-block features are thus processed first by their own maps, and cross-objective information interaction happens uniformly in the subsequent shared trunk. The shared trunk aggregates these latent representations into a scalar score, $g_{\theta}(\Delta) = \bar{T}(\Phi_{\oplus}(\Delta))$ with $\bar{T}(h) = \bar{w}^T \rho(\bar{V}h + c) + b$, whose linear maps are spectrally normalized in the same way so that $\text{Lip}(\bar{T}) \leq 1$. Here g_{θ} denotes the raw critic output before calibration; the score f_{θ} used in the loss and the policy reward is identical to g_{θ} before the scale calibration and afterwards differs by one fixed global scaling factor. All affine layers keep their bias terms; biases shift activation boundaries but cancel in score differences and therefore do not alter the Lipschitz bound.

The overall property of this block structure follows directly from a distance relation. Because the latent representations of the semantic blocks occupy non-overlapping coordinates, for any two differentials Δ and Δ' ,

$$\begin{aligned} \|\Phi_{\oplus}(\Delta) - \Phi_{\oplus}(\Delta')\|_2^2 &= \sum_{g=1}^G \|\psi_g(\Delta_g) - \psi_g(\Delta'_g)\|_2^2 \\ &\leq \sum_{g=1}^G \|\Delta_g - \Delta'_g\|_2^2 = \|\Delta - \Delta'\|_2^2, \end{aligned} \quad (8)$$

so the entire map from the raw differential to the concatenated latent representation remains 1-Lipschitz. Each semantic block first passes through its own independent constrained map, and the shared trunk then learns the relations among objectives; how sensitive the critic should actually be to each objective is learned by the shared trunk from the residual distribution observed during training.

Combining the non-expansive latent maps with the Lipschitz bound of the shared trunk yields, for the raw critic, $|g_{\theta}(\Delta) - g_{\theta}(\Delta')| \leq \|\Delta - \Delta'\|_2$. In motion tracking, perfect imitation always corresponds to the zero differential $\Delta = 0$, the anchor; taking $\Delta' = 0$ gives

$$|g_{\theta}(0) - g_{\theta}(\Delta)| \leq \|\Delta\|_2. \quad (9)$$

This bound—the critic’s geometry, i.e. how fast the score may change with the input residual—is thus a direct consequence of the network structure, independent of the motion.

C. Anchor Calibration: Scale in Closed Form

The bound (9) limits how fast the critic score may change, but not the numerical range the trained critic actually uses; that range is decided jointly by the optimization process and the residual distribution. It directly affects the reward received by the policy. DARE converts the critic score into the policy reward through $r^{\text{DARE}} = \beta \text{softplus}(f_{\theta})$, so the rate of change of the reward with respect to the score is

$$\frac{\partial r^{\text{DARE}}}{\partial f_{\theta}} = \beta \sigma(f_{\theta}). \quad (10)$$

When f_θ is very negative, score changes bring only tiny reward differences; around zero the reward responds markedly to score changes; for large f_θ the slope approaches β and softplus becomes nearly linear. The critic therefore needs not only a suitable sensitivity to residuals but also scores that fall inside the range where the reward transform is sufficiently discriminative.

This scale can be read from the adversarial loss itself. Let the anchor score be $f_\theta(0)$. The residuals produced by the policy follow q_π , and the critic’s mean ideality probability over these residuals is $\mathbb{E}_{\Delta \sim q_\pi} \sigma(f_\theta(\Delta))$. Mapping this mean probability back to the logit space defines the representative score of the typical policy residual, $\bar{f}_\theta = \sigma^{-1}(\mathbb{E}_{\Delta \sim q_\pi} \sigma(f_\theta(\Delta)))$. The balanced Bernoulli adversarial loss is stationary with respect to a global output shift when $\mathbb{E}_{q_\pi} \sigma(f_\theta(\Delta)) = \sigma(-f_\theta(0))$, hence $\bar{f}_\theta = -f_\theta(0)$. Writing $M = f_\theta(0) - \bar{f}_\theta$ for the score gap between the anchor and the typical residual then gives

$$f_\theta(0) = +\frac{M}{2}, \quad \bar{f}_\theta = -\frac{M}{2}. \quad (11)$$

At the reference placement induced by the balanced loss, the anchor score and the typical policy residual score are symmetric about zero, and M measures their separation in logit space.

This gap directly determines the operating regime of the reward signal. When M is too large, typical residuals fall into the low-slope region of the reward transform, so improvements in poor tracking produce little additional reward. When M is too small, anchor and residual scores cluster around zero and the reward differences among tracking qualities are compressed. A useful score scale should therefore place their separation across the main response region of the sigmoid.

A sigmoid has no strict transition boundary, so the operating region needs an operational width. A common choice is the 20–80% span of the sigmoid, for which the corresponding logit width is

$$w = \sigma^{-1}(0.8) - \sigma^{-1}(0.2) = \ln 4 - (-\ln 4) = \ln 16 \approx 2.77. \quad (12)$$

DARE adopts $w = \ln 16$ as the critic’s target operating gap. Under the symmetric reference placement, this width corresponds to $+w/2$ and $-w/2$, whose sigmoid values are 0.8 and 0.2, respectively—a confident but not categorical classifier, in line with the common practice of keeping an adversarial critic in a band of moderate confidence [9]. This operating region is fixed by the reward transform before training and is identical across all motions.

The critic obtained in actual training need not naturally use this scale. DARE therefore measures the score gap M_f of the current critic once, after the input normalization stabilizes, and rescales the whole logit uniformly by

$$f_\theta(\Delta) = \kappa g_\theta(\Delta), \quad \kappa = \frac{w}{M_f},$$

$$M_f = g_\theta(0) - \sigma^{-1}(\mathbb{E}_{q_\pi} \sigma(g_\theta(\Delta))) \Big|_{t=t_f}, \quad (13)$$

where g_θ is the raw critic output before calibration and f_θ the calibrated score that enters the adversarial loss and the

policy reward. When the measured gap M_f exceeds the target width w , $\kappa = w/M_f < 1$ and all critic scores are compressed toward zero, bringing an overly wide score gap back into the main response range of the reward transform; conversely, $M_f < w$ gives $\kappa > 1$ and the critic scores expand away from zero, enlarging the reward differences among tracking qualities. Because the typical residual score used in practice is defined through $\sigma^{-1}(\mathbb{E} \sigma(\cdot))$, the goal of the calibration is to move the overall logit scale toward the operating range set by w ; it does not require every post-calibration statistic to land exactly on a prescribed endpoint.

The gap M_f is measured once at the iteration t_f at which the input normalizer $\mathcal{N}$ is fixed. The computation uses the mixture of current-rollout and replayed residuals, consistent with the non-ideal sample distribution used in critic training. Before t_f , $\kappa = 1$: only after the normalizer is fixed do the units and scales of the differential coordinates stop changing, so a measured score gap first acquires a stable numerical meaning then. After the calibration, κ is held fixed, and both the critic loss and the policy reward use $f_\theta = \kappa g_\theta$; the critic continues to train at one scale, and the policy reads its reward at the same scale.

The uniform rescaling preserves the critic’s original score ordering and multiplies the raw score function by the same constant. The pre-calibration geometric bound $|g_\theta(0) - g_\theta(\Delta)| \leq \|\Delta\|_2$ accordingly becomes $|f_\theta(0) - f_\theta(\Delta)| \leq \kappa \|\Delta\|_2$, a Lipschitz constant derived from the data rather than assumed to be one. The target width $w = \ln 16$ is fixed by the reward transform, and M_f is measured automatically from the training data, so all motions share the same calibration rule while the resulting κ varies automatically with each motion’s residual distribution.

D. Training

Training alternates rollout collection, differential construction, critic updates, and PPO updates (Algorithm 1). Critic minibatches draw an equal mixture of current and replay differentials, with the ideal point presented once per minibatch as the positive class; at t_f , before the rewards of that iteration are computed, M_f is measured with the critic in evaluation mode and κ is set by (13). One critic, one reward, one policy, and one calibration rule make up the whole loop.

V. EXPERIMENTS

A. Experimental Setup

All methods are evaluated using the same 28-DoF, 15-body humanoid model adopted in prior physics-based motion-imitation studies, simulated in Isaac Lab at 120 Hz with actions applied at 30 Hz through PD joint-position control [14], [15]. All four methods share the same environment, training budget, motion clips, random-phase initialization, and evaluation protocol. Each policy is trained for 2000 iterations of 8192 environments $\times$ 32 steps (5.24×10^8 samples) with five independent seeds, and is evaluated on 4096 episodes per seed from randomly sampled reference phases. Actor and value networks are two-layer MLPs (1024, 512), and the PPO settings (clip 0.2, GAE 0.95, $\gamma = 0.99$,

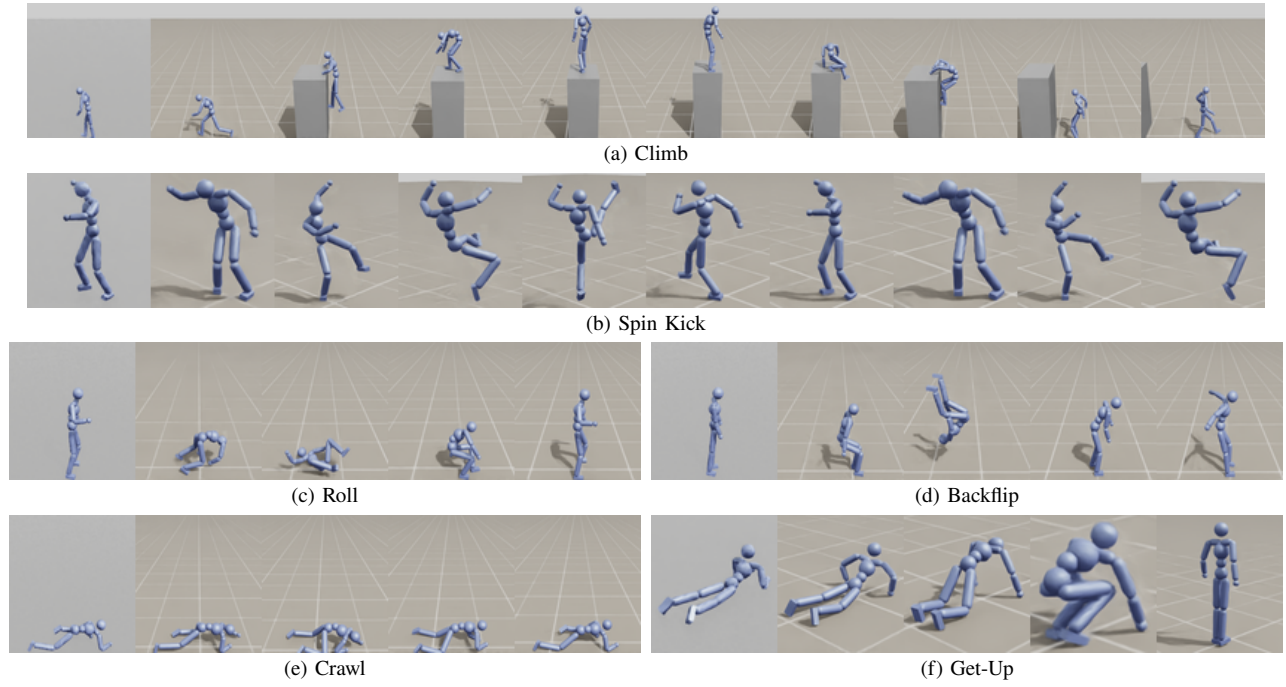


Fig. 1. DARE policies executing the six skills in Isaac Lab using the shared configuration in Table I. Frames are uniformly sampled from each rollout and rendered from a fixed front-view camera. Climb and Spin Kick are shown with ten frames, and the remaining skills with five.

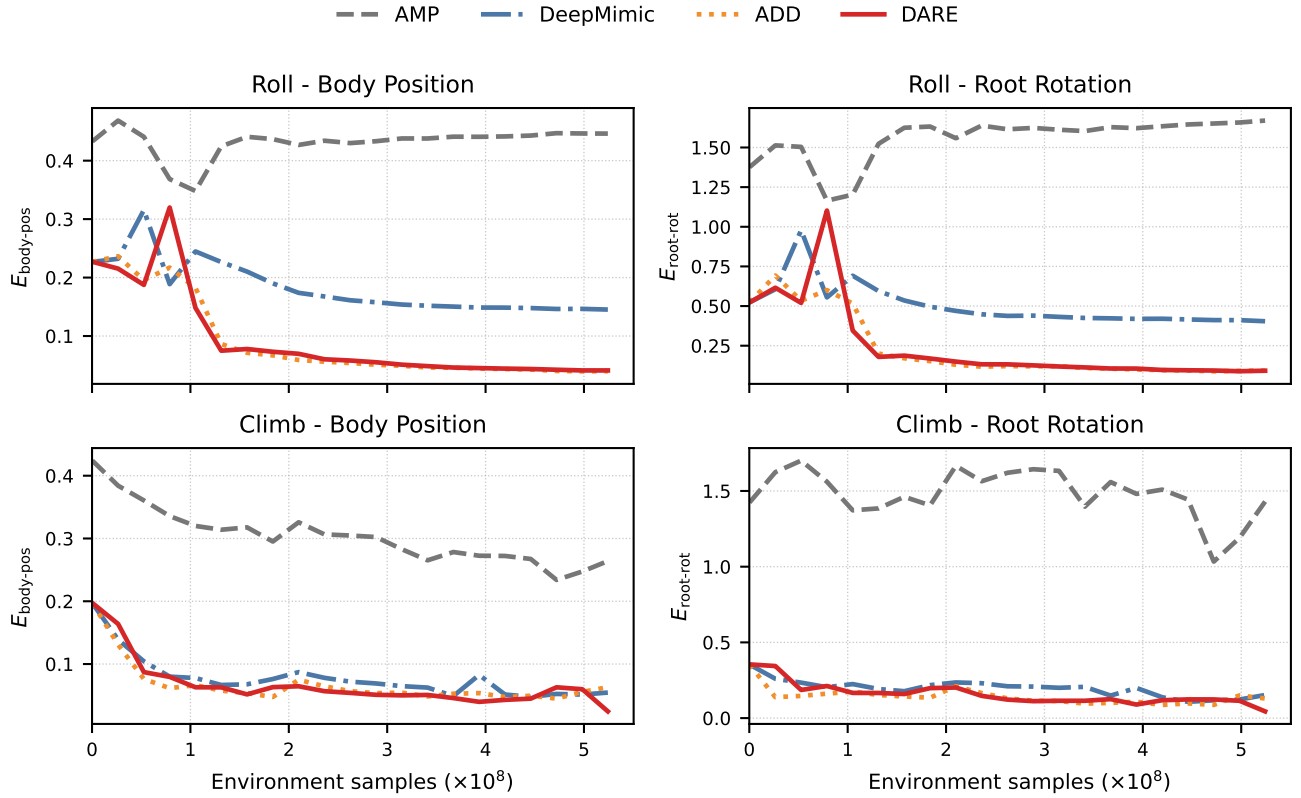


Fig. 2. Learning curves for Roll and Climb under the matched training budget. Curves show the mean over five independent seeds. The four panels report body-position and root-rotation errors versus environment samples, representing articulated translational fidelity and global orientation tracking, respectively. Lower is better.

Algorithm 1 DARE Training

Require: reference motion $\mathcal{M}$, policy π_φ , value V_ω , raw critic g_θ , replay buffer $\mathcal{B}$, normalizer $\mathcal{N}$; $\kappa \leftarrow 1$

- 1: **for** iteration $t = 1, 2, \dots$ **do**
- 2: Collect a synchronized rollout with π_φ from random reference phases; construct Δ_t by (2); add to $\mathcal{B}$
- 3: **if** $\mathcal{N}$ was fixed this iteration **then**
- 4: $\bar{g} \leftarrow \sigma^{-1}(\mathbb{E}_{\mathcal{B}_{\text{Urollout}}} \sigma(g_\theta(\Delta)))$
- 5: $M_f \leftarrow g_\theta(0) - \bar{g}$; $\kappa \leftarrow \ln 16/M_f$
- 6: **end if**
- 7: Compute r_t^{DARE} by (5) with $f_\theta = \kappa g_\theta$
- 8: **for** critic minibatch **do**
- 9: Sample equal numbers of current and replay differentials; update g_θ by one step on (4) with $f_\theta = \kappa g_\theta$
- 10: **end for**
- 11: Update π_φ and V_ω with PPO on r_t^{DARE}
- 12: **end for**

TABLE I

DARE CONFIGURATION SHARED ACROSS ALL SIX SKILLS. THE CALIBRATION RULE IS SHARED; THE RESULTING κ IS DATA-DEPENDENT.

Component	Setting
Differential	single frame, $d = 172$
Semantic factors	$G=7$: root/body pos. (3/45), root/body rot. (6/84), root lin./ang. vel. (3/3), DoF vel. (28)
Direct-sum layer	$m = 146$ per factor (1022 total)
Shared trunk	linear $1022 \rightarrow 512$; output $512 \rightarrow 1$
Activation ρ	ReLU (1-Lipschitz)
Spectral norm.	all linear maps, one power iteration
Normalizer $\mathcal{N}$	per-coordinate mean-abs scale, fixed after 10^8 samples (iter. 382)
Logit calibration	$\kappa = \ln 16/M_f$, once at iter. 382
Critic optimizer	SGD, lr 2.5×10^{-4} , momentum 0.9
Critic minibatch	16384 current + 16384 replay
Replay buffer	2×10^5 differentials
Reward amplitude β	2

SGD with momentum 0.9) are fixed across methods. DARE additionally uses the normalizer and one-shot logit calibration specified in Table I.

The benchmark contains six skills with substantially different motion durations, contact patterns, and dynamical characteristics: Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick. Training termination follows each method’s environment configuration (pose termination is enabled for DeepMimic, ADD, and DARE and disabled for AMP). The evaluation harness accumulates each metric over the executed reference-aligned steps and uses the same aggregation rule for every method; terminated trajectories are not padded with synthetic states. Final policies reach essentially the full prescribed evaluation horizon, so early termination has negligible effect on the reported averages.

We use six complementary measures. The four geometric measures quantify frame-wise imitation fidelity, whereas foot progress and lateral drift capture trajectory-level behavior that is not summarized by instantaneous pose errors. Four describe geometric tracking at two spatial levels: root position and root rotation measure global translation and orientation,

while body position and body rotation measure articulated-body tracking. For an evaluation trajectory with B bodies, where r, p_b and q, q_b denote simulated root, body positions and orientations and hats denote reference values, these quantities are

$$\begin{aligned}
 E_{\text{root-pos}} &= \|\hat{r} - r\|_2, & E_{\text{root-rot}} &= \angle(q, \hat{q}), \\
 E_{\text{body-pos}} &= \frac{1}{B} \sum_{b=1}^B \|(\hat{p}_b - \hat{r}) - (p_b - r)\|_2, \\
 E_{\text{body-rot}} &= \frac{1}{B} \sum_{b=1}^B \angle(q_b, \hat{q}_b).
 \end{aligned} \tag{14}$$

where $\angle(\cdot, \cdot)$ is the shortest angle between normalized quaternions. Geometric quantities are evaluated at each reference-aligned timestep and then averaged over executed timesteps and episodes; the two behavior measures are computed per trajectory and then averaged over episodes. The remaining two measures describe motion-level behavior. Let Δr and $\Delta \hat{r}$ be the simulated and reference root displacements over the evaluated trajectory, and let $\Delta p_j, \Delta \hat{p}_j$ be the corresponding displacements of foot j . The reference-projected progress and transverse drift are

$$\begin{aligned}
 \rho_{\text{foot}} &= \min_j \frac{\Delta p_j^\top \Delta \hat{p}_j}{\|\Delta \hat{p}_j\|_2^2}, \\
 \rho_r &= \frac{\Delta r^\top \Delta \hat{r}}{\|\Delta \hat{r}\|_2^2}, \\
 E_{\text{lat}} &= \frac{\|\Delta r - \rho_r \Delta \hat{r}\|_2}{\|\Delta \hat{r}\|_2}.
 \end{aligned} \tag{15}$$

Foot progress is ideal at one, while lateral drift is minimized at zero. All six quantities are averaged over the evaluation trajectories; lower is better except for foot progress, whose target value is one.

B. Cross-Motion Tracking

Table II asks whether one configuration can serve skills whose contacts, horizons, and velocity profiles differ, and Fig. 1 shows the resulting controllers executing the six skills in Isaac Lab. With one configuration shared across all six skills, DARE achieves the best macro-average on all six measures. It yields the lowest root- and body-level position and rotation errors, the foot-progress ratio closest to one, and the lowest lateral drift. Relative to the strongest baseline for each measure, root-position error, root-rotation error, and lateral drift are reduced by approximately 25%, 21%, and 38%.

C. Learning Dynamics

Figure 2 compares the five-seed mean body-position and root-rotation errors against environment samples for Roll and Climb, which provide contrasting motion durations and contact structures. Across both, the DARE curves converge to lower errors under the same sample budget, complementing the final-checkpoint comparison in Table II.

TABLE II

MACRO-AVERAGE TRACKING AND MOTION-LEVEL BEHAVIOR UNDER THE MATCHED TRAINING AND EVALUATION PROTOCOL. FOR EACH SEED, EACH METRIC IS MACRO-AVERAGED EQUALLY OVER THE SIX SKILLS; ENTRIES REPORT THE MEAN $\pm$ ONE STANDARD DEVIATION ACROSS FIVE SEEDS. LOWER IS BETTER EXCEPT FOR FOOT PROGRESS, WHICH IS IDEAL AT ONE. DARE USES THE SAME CONFIGURATION ON ALL SKILLS.

(a) Geometric tracking				
Metric ↓	AMP	DeepMimic	ADD	DARE
Root Pos. [m]	0.9116 $\pm$ 0.084	0.0703 $\pm$ 0.012	0.0725 $\pm$ 0.010	0.0528$\pm$0.008
Body Pos. [m]	0.2169 $\pm$ 0.031	0.0557 $\pm$ 0.009	0.0363 $\pm$ 0.007	0.0313$\pm$0.006
Root Rot. [rad]	0.7311 $\pm$ 0.076	0.1459 $\pm$ 0.021	0.0673 $\pm$ 0.011	0.0529$\pm$0.009
Body Rot. [rad]	0.8098 $\pm$ 0.069	0.2145 $\pm$ 0.028	0.1358 $\pm$ 0.017	0.1246$\pm$0.013
(b) Motion-level behavior				
Metric	AMP	DeepMimic	ADD	DARE
Foot progress $\rightarrow$ 1	1.1335 $\pm$ 0.046	1.0060 $\pm$ 0.018	1.0109 $\pm$ 0.021	0.9989$\pm$0.012
Lateral drift ↓	0.1127 $\pm$ 0.019	0.0277 $\pm$ 0.007	0.0435 $\pm$ 0.009	0.0171$\pm$0.005

D. Hardware Deployment

[TODO: Report the robot platform and its degrees of freedom, reference retargeting, domain randomization, observation noise, action filtering, hardware control frequency, and the number of trials and successful executions for each skill. State that the shared DARE reward formulation and calibration rule are used without motion-specific modification.]

VI. CONCLUSION

Manual tracking rewards turn the aggregation of heterogeneous imitation objectives into per-motion engineering. Learned differential rewards remove the component-wise weights, but regularization can still couple the critic’s geometric sensitivity to the score range seen by the policy. DARE treats these as separate design quantities: the factorized critic specifies the geometry, and one closed-form calibration sets the score scale without a motion-specific coefficient. Every objective enters through its own spectrally normalized slot, so feature blocks of unequal dimension enter through equal-width, independently normalized representations, and the score change relative to the unique zero differential is bounded by the residual; the critic is then rescaled once, in closed form, to set its operating scale relative to the transition width of the reward transform. Across the six skills, DARE attains the lowest macro-average for all four geometric errors, the foot-progress ratio closest to one, and the smallest lateral drift in Table II. The formulation also provides a direct path toward multi-clip training and other morphologies, where additional objectives can be introduced as semantic blocks.

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